

MOBILE MANIPULATOR

Navigation, manipulation and live occupancy mapping

Measured MATLAB Simulation Report | Ruddrho Mollik | 6 October 2026

1. Executive overview

This project studies the integration of mobile-base navigation, six-axis arm motion, workcell crossing supervision and LiDAR mapping. A two-block pickup and stacking mission is complemented by independent torque-driven controller experiments and a matched uncertainty study. All reported numerical results below come from the uploaded MATLAB data, not from the earlier development runs.

Measure	Result
Mission / modeled collision samples	Completed / 0
Mission duration / base travel	413.05 s / 113.134 m
Minimum modeled clearance / dock error	0.060003 m / 0.039092 m
Controller experiments / uncertainty trials	8 / 20 matched pairs
Execution environment	MATLAB R2024a, PCWIN64

Key finding: computed torque achieves lower nominal tracking error, but PD + gravity performs better under the supplied combined mismatch/noise test and every sampled uncertainty trial. The results demonstrate the importance of evaluating model-dependent control beyond nominal conditions.

Evidence scope: the main-suite log reports completion at 21:34:33 after starting at 21:34:05. These wall-clock timestamps are distinct from the 413.05 s simulated mission duration. Monte Carlo run duration was not logged in that file.

This is an educational simulation study. Ideal sensing, simplified collision geometry and rigid object attachment limit transfer of the numerical performance to real hardware.

2. System model and implementation

Component	Implementation
Mobile base	Planar x, y, yaw; differential side-speed approximation for four rendered wheels.
Six-axis arm	Axes Z-Y-Y-X-Y-X; geometric forward/inverse kinematics and geometric Jacobian.
Path planning	Eight-connected A*, inflated obstacles, mandatory route waypoints and checked quadratic corner blends.
Navigation feedback	Lookahead heading feedback; reduced speed near goal or large heading error; terminal yaw regulation.
AGV supervision	Two verified crossing lanes per long route leg; scripted occupancy interlock and bounded braking.
Manipulation	Quintic joint/Cartesian references, stationary base, acceleration servo and inverse-dynamics torque checks.
Mapping	0.10 m log-odds grid, 5 Hz range updates, incremental odometry and local scan-to-map matching.

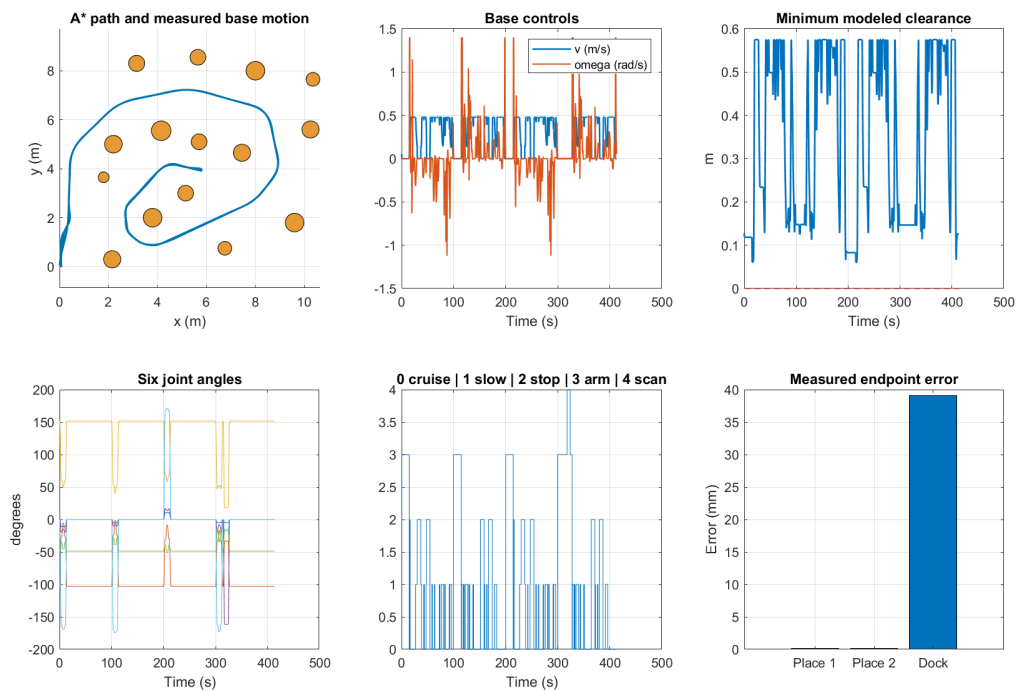
Arm geometry: shoulder height 0.62 m, link lengths 0.62 and 0.52 m, tool offset 0.14 m. These are synthetic model dimensions. The Newton-Euler model includes link mass, approximate inertia, rotor inertia, viscous friction and a point payload.

Base kinematics: $\dot{x} = v \cos(\theta)$, $\dot{y} = v \sin(\theta)$, $\dot{\theta} = \omega$. The torque benchmark integrates $M(q) \ddot{q} + h(q, \dot{q}) = \tau + d(t)$, with a 0.005 s semi-implicit Euler step.

Controller laws: PD + gravity applies $K_p e + K_d \dot{e} + \text{estimated gravity}$. Computed torque applies inverse dynamics to $\ddot{q}_{\text{ref}} + K_p e + K_d \dot{e}$. Both use the same actuator limits: [40, 65, 45, 12, 12, 8] Nm.

Mission and benchmark distinction: mission motion uses an ideal inner acceleration servo. The separate benchmark tests torque-driven dynamics with model mismatch. The successful mission alone does not prove robustness of either benchmark controller.

3. Mission results and station clearance

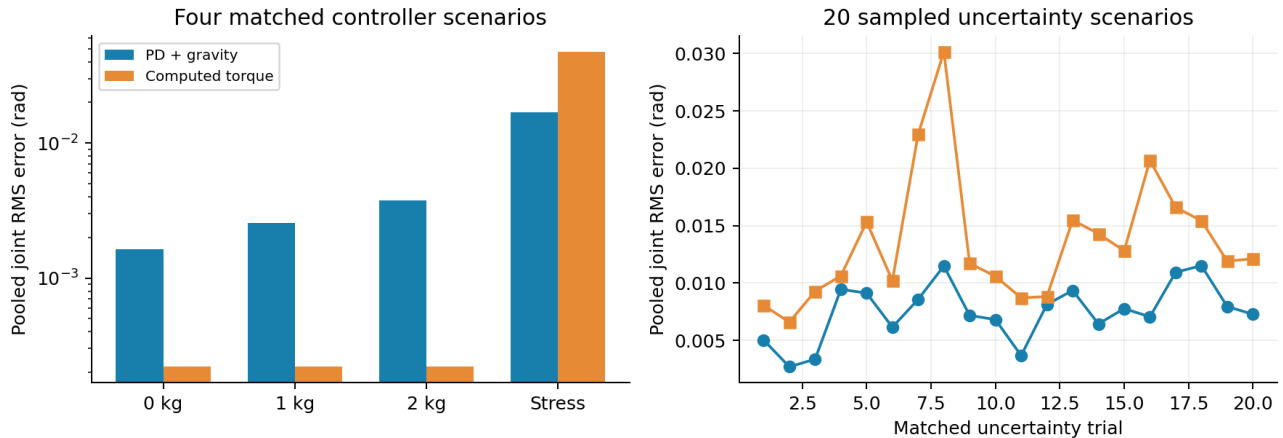


Measurement	Measured value
Navigation nearest-path RMS	0.020103 m
Placement errors	5.817e-07 m; 6.123e-07 m
Dock position / heading errors	0.039092 m / 0.023678 rad
AGV crossings / final scan hold	8 / 6.00 s
Safety-stop mode samples	1140

The supplied log confirms checks for full AGV stops, valid planned path segments, station-pad clearance and scanning only after both releases. The circular base envelope remained at least 0.060003 m from the station pads throughout the checked trajectory. The global minimum modeled clearance was also 0.060003 m.

Pickup/place targets leave the base outside both pads. Manipulation still reaches the blocks with the arm. The micron-scale placement residuals reflect numerical convergence and rigid attachment; they do not represent mechanical repeatability or physical grasp precision.

4. Controller benchmark: nominal versus stress



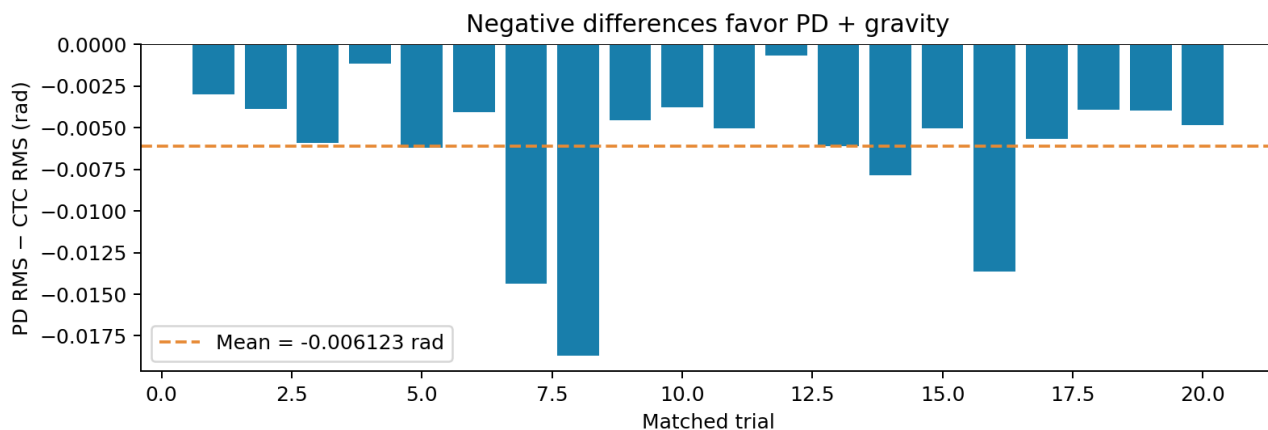
Scenario	PD RMS (rad)	CTC RMS (rad)
Nominal 0 kg	0.00162835	0.00021985
Nominal 1 kg	0.00255107	0.00021985
Nominal 2 kg	0.00373531	0.00021985
Combined stress	0.01688337	0.04747052

The combined stress case uses a 2 kg payload, link mass/inertia scaled by 1.20, position measurement noise with standard deviation 0.003 rad, velocity noise three times that value, and a 2 Nm shoulder disturbance between 2 and 2.5 s. Controllers share the same scenario and random seed.

At nominal 2 kg payload, computed torque reduces pooled RMS by 94.1% relative to PD. In the combined stress case, its RMS is 2.81 times the PD value. All eight trials report zero torque saturation fraction.

RMS pools all six joints and all time samples. Peak error is the largest absolute joint error; final error is the terminal joint-error norm. Under stress, computed torque has a smaller terminal norm (0.026106 versus 0.048580 rad) despite worse RMS and peak error. Conclusions therefore depend on the metric, not just the final pose.

5. Matched Monte Carlo uncertainty study

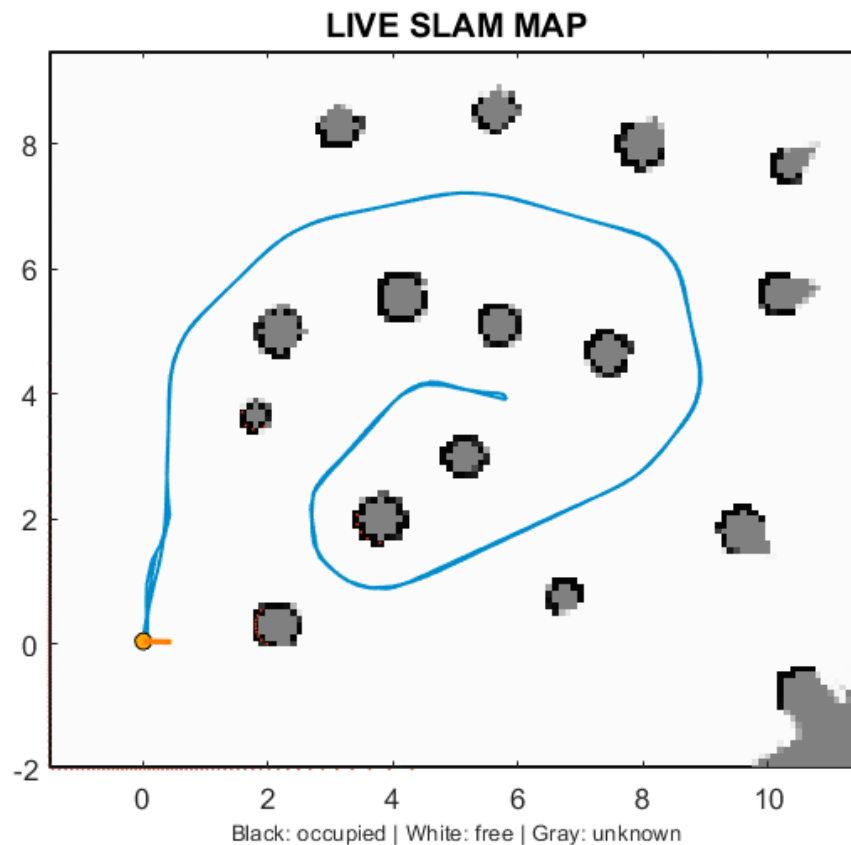


Statistic	Result
Number of scenarios / controller runs	20 / 40
Mean PD RMS	0.00749363 rad
Mean computed-torque RMS	0.01361662 rad
Mean paired difference, PD minus CTC	-0.00612298 rad
Standard error of paired mean	0.00100058 rad
95% paired t interval	[-0.00821723, -0.00402874] rad
Trials with lower PD RMS	20 of 20

The configured sampling ranges are payload 0–2 kg, link mass/inertia multiplier 0.85–1.15 and position-noise standard deviation 0.001–0.005 rad. Each scenario uses matched controller noise seeds. The disturbance pulse is active when the sampled mass scale differs from one, as implemented in `mm_trial`.

The interval is an exploratory paired t interval for the sampled RMS differences. It does not cover other controller tunings, trajectories or uncertainty distributions. The sample count is 20 pairs, not 100 trials per controller. At the supplied gains, PD is consistently less sensitive in this uncertainty sample.

6. Live LiDAR mapping and inspection



The supplied run processed 2066 scans. The log reports 2.83 s of map processing. The map starts unknown and accumulates free-space and endpoint occupancy from range beams. The planner is independent and continues to use known environment geometry.

The maximum estimated-versus-simulated base-position difference is $2.849\text{e-}14$ m, effectively numerical zero. This is expected to be interpreted in the context of ideal simulation odometry and noise-free replay scans. It is not a realistic SLAM accuracy benchmark.

The implementation performs local scan-to-map matching, but no loop closure or global pose-graph optimization. Moving AGV returns can temporarily persist until subsequent free-space observations clear them. The map does not prove perception robustness in unknown environments.

After both boxes are released, a six-second synthetic scan is displayed. Inspection projects known block centroids through a pinhole camera model and checks stack geometry. It does not detect objects or defects from real image pixels.

7. Validation, reproduction and limits

The user MATLAB log records passing forward/inverse kinematics checks, a finite-difference Jacobian comparison, positive-definite mass-matrix checks, gravity/potential consistency, IK workspace rejection, quintic endpoint checks, A* clearance and LiDAR geometry. Mission checks cover eight crossing stops and the final scan sequence; pad-clearance and mapping checks also pass.

This review independently recomputed all eight RMS errors from saved trial arrays and checked the 20 paired differences. The common files in both uploaded archives are identical. The results.zip archive additionally contains Monte Carlo MAT/CSV files, so it is the authoritative complete evidence source for this release.

Task	MATLAB command
Replay supplied mission	clear functions; replay_simulation
Recompute main suite	run_all_results
Recompute uncertainty sample	run_monte_carlo(20)
Export simulation-only video	export_video(30,2)
Package generated evidence	package_results

Open the repository root as MATLAB Current Folder. The main runner preserves a previous results folder under a timestamped name. The simulation and map/dashboard appear in separate windows. A user-supplied edited simulation video and an animated preview are included in assets/. Playback duration is distinct from simulated mission duration; the numerical analysis is unchanged.

Limitations: no wheel slip or mobile-base/arm dynamic coupling; approximate link inertias; no full self-collision or realistic contact physics; conservative scene geometry and sampled arm checks; known/scripted AGV motion; geometric camera inspection; ideal replay sensors; one mission layout and limited controller uncertainty samples.

Next work should introduce noisy odometry and range sensing, evaluate map and localization errors independently, add loop closure if global SLAM is required, compare controller gains fairly across uncertainty levels, and test contact and mobile-base coupling before hardware deployment.

Traceable sources: results/mission.mat; controller_experiments.mat; controller_metrics.csv; monte_carlo.mat and .csv; slam_validation.mat; matlab_run_log.txt; src/mm_trial.m; src/mm_slam_step.m; run_project.m. See docs/EVIDENCE.md for SHA-256 hashes and docs/MODEL_AND_CONTROL.md for the mathematical implementation.